A SPECTRAL REDUCTION OF THE RIEMANN HYPOTHESIS VIA ZETA SPECTRAL TRIPLES
WITH A FULL RECORD OF THE SIXTY-PHASE PROGRAM THAT LED TO IT

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Abstract. We do not claim a proof of the Riemann Hypothesis (RH). What we do is *localize* it: we reduce RH to a single explicit, phase-sensitive positivity statement, we prove every surrounding link in the chain that leads to that statement, we record every dead end of a sixty-phase program together with its precise obstruction, and we build — on the Connes–Consani–Moscovici (CCM) framework of zeta spectral triples — the prolate-operator scaffolding that surrounds the residual wall. The architecture is a ladder of twelve steps: the first ten are proved or constructed-and-validated, they pin the truth of RH to the positivity of the ground-state eigenvalue $\varepsilon_0(\lambda)$ of a localized Weil quadratic form A_λ, and the last two identify that positivity as the one genuinely RH-hard residue — *the wall* — reached identically from primes, zeros, the prolate operator, and de Branges theory. A novel contribution of the present version is the isolation of a single new lemma (the CCM shorted Dirichlet quotient, Lemma 3.9) on which the uniformity of the structure rests; we show it is a genuinely new theorem about the Weil form, not a consequence of the existing prolate literature, and we report how far its proof has been carried. All statements carry a status mark: **(PROVED)** (rigorous proof or citation that holds); **(COMPUTED, PROOF PENDING)** (high precision, proof pending); **(NO-GO)** (tried and refuted, with reason); **(OPEN)** (open but not RH-hard); **(WALL = RH)** (logically equivalent to RH).

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1. Introduction: the program from Phase 1 to Phase 60

The aim has never been to *prove* RH by frontal assault, but to *localize* it: to reduce RH to a single, explicit, finite, phase-sensitive statement, with every surrounding link proved, every dead end recorded with its obstruction, and the genuinely hard residue named. Sixty phases produced (i) several rigorous unconditional results, (ii) a complete catalogue of what does not work and why, (iii) a precise identification — from seven independent directions — of a single master wall, and (iv) the present reduction onto the Connes–Consani–Moscovici (CCM) framework, the only candidate that survives the structural collapse.

1.1. **The governing thesis.** The opening survey (Phase 1) established the organizing thesis: *every analytic front on RH reduces to a positivity statement* — Weil's explicit-formula functional ≥ 0, equivalently Li's $\lambda_n \geq 0$, Bombieri–Lagarias, the reality of the zeros. Any honest attack must either (a) prove a positivity no one can prove (= RH), or (b) be circular, or (c) live in the wrong region. The program's value is in classifying which of (a)–(c) each attempt falls into.

1.2. **Arc A — the ω-class / resonance program (Phases 1–6).** Arc A attacked the Lindelöf side via the ω-class decomposition (organizing n by $\omega(n)$).

- **P2.** $B(N) \leq N^{\varepsilon} \Rightarrow$ Lindelöf. **(NO-GO)** (NG-A1): the target is stronger than Lindelöf and requires it as a hypothesis — circular.
- **P3.** No power-law growth in the resonance sums; an unconditional onset bound $N^* \gtrsim 10^{14}$. A genuine, if negative, quantitative result.
- **P4.** The *coefficient architecture*, not the zero locations, is the discriminant between ζ and its RH-violating cousins; a parity theorem for the Liouville function organizes the even/odd ω-classes. **(PROVED)** (structural).
- **P5–P6.** A putative "phase transition" at $N_c \approx 1.18 \cdot 10^5$ was **(NO-GO)** (NG-A6) — refuted on canonical re-verification, honestly reframed. The lesson: the resonance statistic r is conditioning-dependent (mean zero), so any unconditional cross-correlation claim dies by Montgomery–Vaughan (NG-A3).

Arc A also ruled out, with reasons, the parity-problem sieve (NG-A4), TDA as a detector (NG-A7), the one-sided Gram artifact $\lambda_{\min} \approx 0.866$ (NG-A8), and the criteria-as-paths family Li / Jensen–GORZ / Báez-Duarte (NG-A5) — reformulations of RH, not routes to it.

1.3. **Arc B — the inverse operator and the rigorous Weil detector (Phase 7).** Arc B produced the program's strongest *unconditional* results. The object is the localized Weil quadratic form $Q = M_{\text{zeros}} - M_{\text{arith}}$ in a Hermite–Gauss basis at height T_0, width σ, dimension J.

- **Front A (PROVED).** An off-line zero forces $\lambda_{\min}(Q) < 0$ with explicit δ^2 constants ($\delta =$ distance from the critical line). The form sees RH violations.
- **Front B (PROVED) (unconditional, PNT only).** The truncation floor obeys $\eta(X) \leq A\, e^{-\alpha(\log X)^2}$: the finite-$X$ error is super-polynomially small, no RH.
- **Front C.** The detector is technical; the *uniform* passage $Q_X \to Q_\infty$ is RH-equivalent — the first clean appearance of the master wall.

The compression question (0A2) — is $Q = P_J \mathcal{T} P_J$ a faithful compression of the Weil operator $\mathcal{T}$? — found both scenarios supported, leaving the named wall (LB) : $\inf \operatorname{spec}(\mathcal{T}) \geq 0 \iff \lambda_{\min}(Q_\infty) \geq 0\ \forall f =$ Weil positivity = RH (if 0A2 holds). **(WALL = RH)**

1.4. **The b_j lower-bound program and local–global (Phases 22–28).**

- **Phases 22–23 (Paley–Wiener barrier).** Arithmetic information from $\operatorname{Re} s > 1$ provably does not propagate into $(1/2, 1)$. This is the central wall **MW-2**.

- **Phases 24–25 (fifteen-mechanism audit).** Fifteen classical routes to a *lower* bound on b_j, each refuted with its failure mode **(NO-GO)** (NG-B1...B8 and more): Korobov–Vinogradov and de Bruijn–Newman give the wrong direction; Deuring–Heilbronn lives near $\operatorname{Re} s = 1$; Beurling–Nyman, Turán-for-ξ_0, Li/Krein–LP are circular; Baker is the wrong category; Montgomery pair correlation and the large sieve carry no off-line signal. Universal cause: MW-2.
- **Phase 26.** Explicit eigenvectors $x^{b_j+i\gamma_j} \notin L^2(C_{\mathbb{Q}})$ **(NO-GO)** (NG-C3): the norm diverges; they are distributions.
- **Phase 27 (local–global).** $Q = Q_\infty \prod_p Q_p$ fails on two walls **(NO-GO)** (NG-C1): the zero side does not factor over primes (no Frobenius-at-p for $\zeta/\mathbb{Q}$, **MW-3**); Hasse–Minkowski fails for ∞-dim Hermitian forms (**MW-5**). Its obstruction equals Phase 26's (NG-C2).

1.5. **The broad exploration (Phases 32–59).** Phases 32–59 swept the structural landscape: CCM absolute geometry (32), de Bruijn–Newman (33), five fronts and ABC forms (35–36), a physics/dynamics import (37, 53), the G_1/G_2 interface (39–41), Hodge-theoretic and foliated $\operatorname{Spec}\mathbb{Z}$ dynamics (42–43), creative rupture / new mathematics (44), a Fredholm route (45), "crossing the wall" (49), a Diophantine thread (50), functor and homotopy-inertia constructions (51–52), index dynamics (54), the "two arrows" / "two towers" (55–56), and the total audits and closure (46–48, 58–59). The honest verdict: no genuinely new structure escaped the collapse — which made the pivot to CCM forced, not fashionable.

1.6. **The seven-fold wall and the meta-search collapse theorem.** A single obstruction reappears under seven names: **MW-1** Weil positivity; **MW-2** arithmetic propagation ($\operatorname{Re} s > 1 \not\to (1/2, 1)$, central); **MW-3** Hasse–Minkowski in ∞-dim; **MW-4** the Hilbert–Pólya *wrong-sign* wall — every unconditional positivity tool gives a lower bound, RH needs an upper; **MW-5** Connes–Consani arithmetic-site; **MW-6** uniform spectral gap, itself = RH. An adversarial novel-structure search, literature-merged and attacked to destruction, proved the collapse theorem: *every RH-directed structure reduces to CAP (positivity), SURF (Connes–Consani geometry, RH-equivalent), or F (statistics).* The literature reproduces it — Kurasov–Sarnak Fourier quasicrystals (reality of support = Lee–Yang = stable polynomial = Hermite positivity = CAP), Deninger (= CAP), Nyman–Beurling/Li (= C), de Branges/Sonine (= Hermite–Biehler reality), Berry–Keating xp (= Hilbert–Pólya), CFKRS/RMT (= F). Two escape candidates destroyed: **(A)** crystalline rigidity = Lee–Yang = positivity; **(B)** functional-equation edge-anchor = the Weil/de Branges form = CAP. The center is reached by definiteness, never by an anchor.

1.7. **The forced pivot: CCM spectral triples (Phase 60).** The collapse leaves one live, non-circular candidate. F cannot decide reality; CAP is the wall. SURF — the CCM spectral triples — is distinguished by **MW-4 inverted**: the operators are self-adjoint *by construction*, so the spectrum is real automatically and the open problem is operator convergence $H_x \to H_\infty$, *not* the sign. This converts the wrong-sign wall into an approximation question — the one place the master obstruction does not obviously apply. Phase 60 implements CCM faithfully and pushes the convergence/positivity question as far as it goes, yielding the reduction chain of §3.

2. Paths explored and no-go results

2.1. **Arc A (ω-class / resonance).**

Tag	Attempt	Why it fails
NG-A1	$B(N) \le N^{\varepsilon} \Rightarrow$ Lindelöf	Target stronger than Lindelöf; circular.
NG-A2	Extrapolate $\alpha_B \to 0$	$\alpha_B \approx 0.42$ vs threshold < 0.31; never crosses.

Tag	Attempt	Why it fails
NG-A3	Unconditional cross-covariance as signal	Montgomery–Vaughan: ≈ 0 unconditionally.
NG-A4	Sieve to separate ω-parity	Parity problem blocks all standard sieves.
NG-A5	Li / Jensen–GORZ / Báez-Duarte as a path	Reformulations of RH; circular.
NG-A6	Phase transition at N_c	Refuted on re-verification; no transition.
NG-A7	TDA as a zero detector	H_0 is δ^2-insensitive; H_1 no signal.
NG-A8	$\lambda_{\min} \approx 0.866$ as positivity	One-sided Gram trivially positive.

2.2. **The b_j lower-bound mechanisms (Phases 24–25).**

Tag	Attempt	Why it fails
NG-B1	Fifteen classical mechanisms en bloc	MW-2: $\operatorname{Re} s > 1$ info does not reach $(1/2, 1)$.
NG-B2	Korobov–Vinogradov zero-free region	Gives $b_j \leq 1/2 - c/(\log \gamma_j)^{2/3}$ — upper bound.
NG-B3	Deuring–Heilbronn repulsion	Operates near $\operatorname{Re} s = 1$; no anti-Siegel off the line.
NG-B4	Beurling–Nyman distance	$d > F(b_j)$ from below $\iff$ bounding b_j — circular.
NG-B5	de Bruijn–Newman $\Lambda \geq b_j^2/2$	Inverted: gives $b_j \leq \sqrt{2\Lambda}$, trivial.
NG-B6	Turán inequalities for ξ_0	Satisfied for all $b_j > 0$ — no constraint.
NG-B7	Baker (linear forms in logs)	Applies to algebraic numbers; γ_j transcendental.
NG-B8	Montgomery pair correlation	On-line spacings only; no off-line signal.

2.3. **Local–global and distributional (Phases 26–27).**

Tag	Attempt	Why it fails
NG-C1	$Q = Q_\infty \prod_p Q_p$ (Hasse–Minkowski)	Zero side does not factor over primes (MW-3); H–M fails ∞-dim (MW-5).
NG-C2	Phase 27 as alternative to 26	Same obstruction, two languages.
NG-C3	Eigenvectors $x^{b_j + i\gamma_j} \in L^2$	Norm diverges; distributions, $Q(f_j, f_j) = \infty$.

2.4. **The structural walls and the meta-search.** *MW*-1 Weil positivity; *MW*-2 arithmetic propagation (central); *MW*-3 Hasse–Minkowski ∞-dim; *MW*-4 Hilbert–Pólya wrong sign; *MW*-5 Connes–Consani arithmetic site; *MW*-6 uniform spectral gap (= RH) — the same wall from seven directions. Every RH-directed structure (corpus or literature) collapses to CAP (positivity), SURF (Connes–Consani, RH-equivalent), or F (statistics); the two purpose-built escapes (crystalline rigidity = Lee–Yang; functional-equation anchor = Weil/de Branges) are destroyed.

2.5. **Phase-60 no-gos (the CCM scaffolding).**

Path	**Why it fails**
$H1$ proves QW PSD via Perron–Frobenius	PF gives ground-state structure, not positive-definiteness. PSD = RH.
Limit operator constant $c = -\psi''(1/4)/8$	Archimedean curvature subdominant by $\sim \lambda \log^2 \lambda$; refuted by $(2\lambda)^{-1} D_\lambda(s/\log\lambda) \to \cos 2s$.
Hankel edge model, $\gamma \to 1$, scale 2λ	False premise $\lvert\varepsilon_0\rvert \sim 2\lambda$; in fact $\sim 2.4 \cdot 10^{-48}$ at $\lambda^2{=}11$.
Dirichlet box / standard Agmon at e^{-L}	Box gives $1/\log^2\lambda$, tunneling $\varepsilon_1/\varepsilon_0 \to \infty$; neither $e^{-L}(k{+}1)^2$.
Magnitude methods for $\operatorname{sign} \varepsilon_0$	Marginality $\sqrt{2}/\pi$; depends only on $\{\lvert\Lambda(n)\rvert\}$; DH same magnitudes, $\varepsilon_0 < 0$.
Frequency law $T_p = \pi/\log p$ for δa_n	Confirmed only for $p = 2$; fits reject $2 \log p$ for $p \geq 3$.
Project J_S out for Sonin	ξ_S cyclic $\Rightarrow$ complement $\{0\}$; Sonin in position space.
$E_S = 1/\Lambda_S$ as de Branges function	Correct modulus, not Hermite–Biehler; needs the functional equation.

Convergent lesson. Every route to the zeros — ω-resonance, the localized detector, b_j bounds, local–global, the CCM prolate, the de Branges/Sonin side — terminates at the same place: the reality of the γ_n, equivalently Weil positivity, equivalently RH. The sign is a *phase* phenomenon, immune to magnitude estimates, to local-operator models, and to any structure that does not encode the zeros by construction.

3. The proof chain

3.1. **Architecture of the proof: where RH would be proved.** Before the technical steps, we explain the shape of the argument as a whole, so that the role of each step — why it is there, what it receives from the step before, and what it hands to the step after — is visible in advance. The reader who keeps the following six-sentence skeleton in mind will not get lost in the lemmas.

The strategy is to convert RH from a statement about the location of zeros into a statement about the sign of a single number, and then to prove every link except that sign. The object that carries the conversion is a one-parameter family of quadratic forms A_λ, $\lambda > 1$, built from the Weil explicit formula and localized to band-limited functions of bandwidth λ. Its construction and meaning are **Step 1**: A_λ is simultaneously (a) a concrete convolution operator with an explicit symbol $a(t) = \widetilde{\Omega}(t) - D_\lambda(t)$, and (b), by the Weil explicit formula, a sum over the zeros ρ of ζ — so its positivity is *equivalent* to the zeros being real. This double identity is the hinge of everything: the analytic side (the symbol) is computable, the arithmetic side (the zeros) is what we want to control.

The chain then proceeds as follows. **Step 2** analyses the bottom of the spectrum of A_λ: it shows the lowest eigenvalue $\varepsilon_0(\lambda)$ is simple with a clean eigenfunction, and — this is the new and delicate part — that the *gap* above it does not collapse as $\lambda \to \infty$, so that the family A_λ has a stable limiting shape. **Step 3** proves that, because A_λ is a compression of a convolution (difference-kernel) operator, the entire function $\hat{\xi}_\lambda$ attached to its ground state has only real zeros at each finite λ. **Steps 4–5** are the reduction proper: the finite-λ object approximates the completed zeta function Ξ with explicit error, and real-rootedness passes to the limit *if and only if* a single localized quantity $\varepsilon_{\mathrm{loc}}(\lambda)$ decays faster than $\lambda^{-1/2}$. At this point RH has been turned into an inequality about one number. **Steps 6–7** certify that the reduction is not circular and not vacuous: a known RH-false model (Davenport–Heilbronn) is correctly *rejected* by the same machinery, and the classical

Hurwitz/Laguerre–Pólya theory supplies the limiting logic. **Steps 8–10** identify the limit operator inside the CCM prolate/zeta-spectral- triple framework and reconstruct, as exact identities, the arithmetic content (periodic-orbit / Gutzwiller sums over primes) that any correct operator must carry; these steps are RH-neutral scaffolding — they make the operator concrete without assuming the answer.

This is where RH would be proved. After Step 5, RH is equivalent to $\varepsilon_{\rm loc}(\lambda) = o(\lambda^{-1/2})$, equivalently (Step 12.1) to the positivity

$$(P): \qquad \varepsilon_0(\lambda) \geq 0 \quad \text{for all } \lambda > 1.$$

Every step 1–10 is in place to make (P) a faithful, non-circular, phase-sensitive restatement of RH and to surround it with an explicit operator. **Steps 11–12 are the wall**: they show that (P) is exactly as hard as RH itself — it is reached identically from the primes, from the zeros, from the Sonin space of the prolate operator, and from de Branges canonical-system positivity, and the marginality theorem (Step 12.2) proves that no *magnitude* method can decide its sign, because the sign is a *phase* invariant. A proof of RH along this route is therefore precisely a proof of (P): one must show the localized Weil form never goes negative, and the program's contribution is to have reduced RH to that one statement with everything around it proved, and to have identified — in Lemma 3.9 — the single new analytic fact (the arithmetic deformation of the spectral ladder) on which the *stability* of the whole picture depends.

Throughout, a status mark records the epistemic status of each claim (see the abstract), so the boundary between what is proved and what is the wall is never blurred.

Notation. $\lambda > 1$, $L = 2\log\lambda$, $\Omega = \log\lambda$; Λ von Mangoldt, $\psi(X) = \sum_{n\leq X}\Lambda(n)$; γ_ρ the ordinates of the nontrivial zeros, $T^* = 2\pi\lambda^2$; $\widetilde{\Omega}(t) = -\log\pi + \operatorname{Re}\psi_\Gamma(\frac{1}{4} + \frac{it}{2})$ the archimedean symbol, $D_\lambda(t) = \sum_{n\leq\lambda^2}\Lambda(n)n^{-1/2}\cos(t\log n)$ the prime symbol; E_λ the band-limited functions of exponential type Ω, spectral support in $[-L/2, L/2]$.

3.2. **Step 1 — The Weil quadratic form and its symbol (PROVED).** *Purpose, and why it comes first.* Everything downstream is an analysis of one object, so we must first build it and establish the one property that gives it meaning: that its positivity is equivalent to RH. The Weil explicit formula expresses a sum over the zeros of ζ as a sum of an archimedean term and a sum over prime powers. Read as a quadratic form in a test function f, it becomes a Hermitian form A_λ whose value is, on one hand, a transparent analytic expression and, on the other hand, literally $\sum_\rho |\hat{f}(\gamma_\rho)|^2$ when the zeros are real. We localize to $f \in E_\lambda$ (band-limited, bandwidth λ) so that only finitely many primes ($n \leq \lambda^2$) enter and the form is a genuine compact-interval operator. Concretely, for $f \in E_\lambda$,

$$A_\lambda(f,f) = \int \widetilde{\Omega}(t)\,|\hat{f}(t)|^2\,dt - \int D_\lambda(t)\,|\hat{f}(t)|^2\,dt,$$

the difference of the archimedean symbol $\widetilde{\Omega}$ and the prime symbol D_λ tested against the power spectrum $|\hat{f}|^2$. The three lemmas below establish, in turn, the three faces of A_λ we will use later: its kernel (so that Step 3's convolution theorem applies), its Fourier symbol (so that Steps 2 and 12 can do spectral perturbation theory on $a(t)$), and its Weil identity (the RH-equivalence that justifies the entire enterprise).

Lemma 3.1 (Schwartz kernel)**.** *A_λ is the compression to $[-L/2, L/2]$ of convolution by $D_\zeta(|x-y|)$, where $D_\zeta(u) = w_{\rm arch}(u) - \sum_{n\leq\lambda^2}\Lambda(n)n^{-1/2}\delta(u - \log n)$.*

Proof. From $Q(f,g) = \int_{-L}^{L}(g\star f)(y)D_\zeta(|y|)dy$, $(g\star f)(y) = \int\overline{g(x)}f(x+y)dx$, $h(u) := D_\zeta(|u|)$ even, change variables $(x,y)\mapsto(x, y'{=}x{+}y)$: $Q(f,g) = \iint\overline{g(x)}f(y')h(y'{-}x)dx\,dy' = \int\overline{g(x)}\big(\int k(x,y')f(y')dy'\big)dx$, $k(x,y') = D_\zeta(|x-y'|)$. Restriction gives the compression. $\square$

Lemma 3.2 (Fourier multiplier)**.** *$\hat{h}(t) = 2\int_0^\infty D_\zeta(u)\cos(tu)du = \widetilde{\Omega}(t) - D_\lambda(t) =: a(t)$.*

Proof. The prime part gives $D_\lambda(t)$; the archimedean part $2\int_0^\infty w_{\mathrm{arch}}(u)\cos(tu)du = \widetilde{\Omega}(t)$ by the explicit-formula relation. ☐

Lemma 3.3 (Weil identity)**.** *$A_\lambda(f,f) = \sum_\rho \hat{f}(\gamma_\rho)\overline{\hat{f}(\overline{\gamma_\rho})}$; under RH, $= \sum_\rho |\hat{f}(\gamma_\rho)|^2 \geq 0$.*

Proof. Weil's explicit formula on $g \star f$, support $2\Omega = L$ truncating primes to $n \leq \lambda^2$; archimedean and pole terms give $w_{\mathrm{arch}}, \mathcal{P}$; the zero sum is indefinite absent reality of γ_ρ. ☐

Finally we record the form in *carré-du-champ* (energy) coordinates, which is the representation Step 2 needs: symmetrizing the kernel splits A_λ into a mass term and two non-negative Dirichlet ("energy") forms,

$$A_\lambda(f,f) = \underbrace{(\widetilde{\Omega}(0) - P_\lambda)\|f\|^2}_{\text{mass}} + \underbrace{\mathcal{E}_\theta(f)}_{\text{archimedean jumps}} + \tfrac{1}{2}\underbrace{\mathcal{E}_{\mathrm{prime}}(f)}_{\text{prime jumps}}, \qquad \mathcal{E}_\theta, \mathcal{E}_{\mathrm{prime}} \geq 0.$$

The point of this rewriting is that it separates what can make A_λ negative (the scalar mass coefficient $\widetilde{\Omega}(0) - P_\lambda$) from what is manifestly positive (the two energy forms). Step 2 exploits exactly this separation. **(PROVED)**

3.3. **Step 2 — Structure of the ground state (PROVED) (λ finite).** *What Step 2 must deliver, and why.* For the reduction in Step 5 to pass to the limit we need two things about the bottom of the spectrum of A_λ: that the ground state is *clean* (simple, isolated, with a sign-definite eigenfunction, so that the entire function $\hat{\xi}_\lambda$ of Step 3 is well defined) and that it is *stable* (the spectral gap above ε_0 does not close as $\lambda \to \infty$, so that the limit $\lambda \to \infty$ in Step 5 does not degenerate). The first is a soft Perron–Frobenius argument and is unconditional; it uses only the positivity of the energy forms isolated in Step 1. The second is the genuinely hard, genuinely new content of this paper, and occupies Lemmas 3.7–3.9 and Theorem 3.10 below. We begin with the positivity structure that makes the semigroup of A_λ positivity-preserving.

Lemma 3.4 (Lévy–Khintchine positivity)**.** *$M_\theta \succeq \widetilde{\Omega}(0)I$, and $M_\theta - \widetilde{\Omega}(0)I$ is a jump Dirichlet form; its semigroup is positivity-preserving.*

Proof. $\widetilde{\Omega}(t) - \widetilde{\Omega}(0) = \int_0^\infty (1-\cos t\xi)d\nu(\xi)$, $d\nu = 2e^{-\xi/2}(1-e^{-2\xi})^{-1}d\xi \geq 0$ (verified $\int \min(1,\xi^2)d\nu < \infty$). Hence $\langle f, M_\theta f\rangle - \widetilde{\Omega}(0)\|f\|^2 = \frac{1}{2}\iint |f(w) - f(w+\xi)|^2 d\nu\, dw \geq 0$. ☐

Theorem 3.5. *At each finite λ, $\varepsilon_0(\lambda)$ is simple, isolated, with even constant-sign eigenvector ξ.*

Proof. $A_\lambda = M_\theta - V$, $V = \sum \Lambda(n)/\sqrt{n}\, T_{\log n}$. e^{-tM_θ} and (since $\Lambda \geq 0$) e^{tV} positivity-preserving; Trotter $\Rightarrow$ Perron–Frobenius $\Rightarrow$ bottom simple, isolated, constant sign; $\iota : x \mapsto L - x$ forces evenness. ☐

Remark 3.6 (uniformity — the rescaled low spectrum). The uniformity $\liminf_\lambda(\varepsilon_1 - \varepsilon_0)/|\varepsilon_0| > 0$ is not delivered by the Perron–Frobenius argument, which is λ-by-λ. With the validated CCM engine (dps 40, primes included) the rescaled low spectrum of A_λ obeys $\varepsilon_k/\varepsilon_0 \to (k+1)^2 = 4, 9, 16, 25, \ldots$, with $\varepsilon_2/\varepsilon_0 \to 9.0$ very cleanly and $\varepsilon_1/\varepsilon_0 \in [3.51, 4.19]$ robustly across $\lambda \in [3, 12]$, $N \in [8, 24]$. This is the *positive* prolate near-radical spectrum (the Slepian χ_m), not the negative Sonin spectrum of Step 11. Two naive routes are refuted (NO-GO): (i) archimedean skeleton + bounded prime perturbation (primes off $\Rightarrow$ M_θ indefinite, $\widetilde{\Omega}(0) \approx -5.37 < 0$, no ladder — the ladder is constitutively arithmetic); (ii) free Dirichlet box (spectrum $(k+1)^2$-like but eigenvectors are not box modes). The correct identification follows.

Strategy for the uniform gap. We must show $\liminf_\lambda(\varepsilon_1 - \varepsilon_0)/|\varepsilon_0| > 0$. The obstacle is that the absolute scale of the low eigenvalues is super-exponentially small ($\rho_\lambda \asymp e^{-cL}$), so any argument that tries to track ε_0 itself is hopeless; we must work with *ratios*, in which the unknown scale cancels. The plan has three moves. First (Lemma 3.7, the Doob identity) we record the exact change of variables that turns the candidate limit operator into a Dirichlet Laplacian, so that "the ratios are m^2" and "the eigenfunctions are Chebyshev" become the same statement. Second (Lemma 3.8, scale-free Parter) we prove the general principle that a Dirichlet-type form has ratios $\to m^2$ *independently of its overall scale* — this is what licenses us to ignore ρ_λ. Third (Lemma 3.9, the CCM shorted Dirichlet quotient) we state the one fact that is genuinely about the explicit arithmetic of the Weil form: that the conditioned (shorted) form on the slow sector *is* a Dirichlet form. Theorem 3.10 then combines the three into the uniform gap, and Corollary 3.11 feeds that gap into Step 3 to make real-rootedness survive the limit. Lemma 3.9 is the only assumed ingredient; the remark after it explains precisely why it is new and how far its proof has been carried.

Lemma 3.7 (Doob identity — PROVED)**.** *For $f = \sin\theta \cdot v$, $\int_0^\pi |f'|^2 - \int_0^\pi |f|^2 = \int_0^\pi |v'|^2 \sin^2\theta$.*

Proof. $f' = \cos\theta\, v + \sin\theta\, v'$; the cross term $\int 2\sin\theta\cos\theta\, vv' = -\int \cos 2\theta\, v^2 = -\int v^2 + 2\int \sin^2\theta\, v^2$ (by parts, boundary terms vanish); collecting with $\cos^2\theta - 1 = -\sin^2\theta$ everything cancels to $\int \sin^2\theta\, v'^2$. Hence the Dirichlet Laplacian (eigenvalues m^2, eigenfunctions $\sin m\theta$), shifted by its ground energy 1 and transformed by $s_1 = \sin\theta$, becomes $L_\infty = -\sin^{-2}\theta\,\partial_\theta(\sin^2\theta\,\partial_\theta)$ (Chebyshev $U_k(\cos\theta) = \sin((k+1)\theta)/\sin\theta$, eigenvalues $k(k+2)$). ∎

Lemma 3.8 (scale-free Parter — PROVED)**.** *If $F_N = \tau_N D_D^* C_N D_D$ with D_D the Dirichlet first-difference and $C_N \to c\,I$ in low frequencies, then $\lambda_m(F_N) = \tau_N c \cdot 4\sin^2(\pi m/2(N+1))(1+o(1))$, so $\lambda_m(F_N)/\lambda_1(F_N) \to m^2$ for fixed m — independent of the scale τ_N (it cancels in the ratio). This is why the super-exponential $\rho_\lambda \asymp e^{-cL}$ need never be computed.*

Lemma 3.9 (CCM shorted Dirichlet quotient — ASSUMED; the genuine residual)**.** *In the CCM objects: let $\mathcal{E}(f)(x) = x^{1/2}\sum_{n>0} f(nx)$ be the conditioning map (Connes–Consani, arXiv:2106.01715, eq. 3.1), let φ_n be the prolate combinations vanishing at* 0*, and let ε_n be the Gram–Schmidt orthonormalization of $\{\mathcal{E}(\varphi_n)\}$ (eq. 3.4), spanning the prolate projection $\Pi(\lambda, k)$ (Def. 3.1). Then for every fixed N, $\rho_\lambda^{-1}\langle\varepsilon_m, QW_\lambda\varepsilon_n\rangle \to m^2\delta_{mn}$ $(1 \le m, n \le N)$, i.e. the shorted/quotient form of QW_λ on the near-radical is the Dirichlet form $-\partial_\theta^2$. This is not a soft consequence of Lemmas 3.7/3.8; it is a genuine theorem about the explicit $\mathcal{E}$, here assumed. Validated: (i) eigenvectors $\varepsilon_0, \varepsilon_1, \varepsilon_2$ are the Dirichlet sine modes (arXiv:2106.01715, Figs. 22–24), matching the true QW_λ eigenfunctions (Figs. 26–37); (ii) Rayleigh quotients $R(U_k) \to k(k+2) = 3, 8, 15$ (exact engine, $\lambda = 7, 10, 13$); (iii) overlaps $0.99, 0.80, 0.51$; (iv) $\varepsilon_k/\varepsilon_0 \to (k+1)^2$ robust in λ, N.*

Status (2026-06-21; head-to-head vs the CCM prolate Jacobi — Connes-validated, his version pending). *The ladder is a property of the Weil form QW_λ,* not *of the bare prolate W_λ, and is not in the CCM prolate papers. The exact CCM Jacobi (arXiv:2310.18423, Prop. 3.5: $b_n = -8n^2 - 2n - \frac{3}{4} + 2\pi\lambda^2(8n+1)$) gives, under Schur complement, the* linear *ladder $8n+1$ (normalized $0, 1, 2, 3, 4$); the engine QW_λ gives the* quadratic *ladder $\to (k+1)^2$. So the quadratic ladder is constitutively arithmetic, living in $QW_\lambda - W_\lambda$ (prime + archimedean terms):* ***2.3.F is a new theorem, not a citation.*** *Route (A) diagnosis (partial, not closed): in the symbol $a(t) = \widetilde{\Omega}(t) - D_\lambda(t)$ the curvature $a''(0) = \widetilde{\Omega}''(0) + M_2$ with $M_2 = \sum \Lambda(n)(\log n)^2/\sqrt{n} \sim 8\lambda\log^2\lambda$ dominates, so $\frac{1}{2}M_2 t^2 \leftrightarrow -\partial_w^2$ is the local Laplacian with arithmetic coefficient. Open piece: the marginality $\varepsilon_0 \approx 0$ does not follow from the bare symbol ($a(0) \approx -17 < 0$); it comes from the regularization WR/near-radical, which also supplies the Dirichlet confinement. Assumed pending Connes' version.*

Theorem 3.10 (uniform gap — PROVED modulo Lemma 3.9)**.** *By Lemma 3.9 and sine-diagonalization (Lemma 3.8), $\rho_\lambda^{-1}\widetilde{F}_{\lambda,N} \to (-\partial_\theta^2)_{\mathrm{Dir}}$, so $\widetilde{\varepsilon}_m/\widetilde{\varepsilon}_1 \to m^2$; subtracting the ground state and setting $m = k+1$, $(\varepsilon_k - \varepsilon_0)/|\varepsilon_0| \to (k+1)^2 - 1 = k(k+2)$, in particular $(\varepsilon_1 - \varepsilon_0)/|\varepsilon_0| \to 3 > 0$. By Lemma 3.7 the limit operator is L_∞ (Chebyshev). Hence $\liminf_\lambda(\varepsilon_1 - \varepsilon_0)/|\varepsilon_0| \ge 3 > 0$: the uniformity of Step 2 holds. (RH-neutral: phrased as $(\varepsilon_k - \varepsilon_0)/|\varepsilon_0|$, it does not decide $\operatorname{sign}\varepsilon_0$.)*

Corollary 3.11 (real-rootedness in the limit, Step 3.2 — PROVED from Thm 3.10)**.** *The uniform gap isolates the ground-state spectral projection uniformly in λ. By Kato stability the projections converge; by Montel the $\hat{\xi}_\lambda$ form a normal family with a non-degenerate limit; by Hurwitz the real-rootedness of each $\hat{\xi}_\lambda$ (Step 3.1, CvS) passes to the limit. Hence Ξ is real-rooted on the relevant range — Step 3 survives $\lambda \to \infty$.* ***(Step 3.2 PROVED modulo Lemma 3.9.)***

3.4. **Step 3 — Reality of the zeros of $\hat{\xi}$ (PROVED).** *Why this step, and what it receives from Step 2.* Step 2 produced, at each finite λ, a clean ground state $\xi = \xi_\lambda$; attached to it is an entire function $\hat{\xi}_\lambda$ (its Fourier transform) whose zeros are the finite-λ analogue of the zeta zeros. The whole reduction will compare these zeros to the true zeros of Ξ. For that comparison to mean anything we must first know that, *at each finite* λ, the $\hat{\xi}_\lambda$ already has all its zeros real — otherwise there would be nothing to pass to the limit. This is the Connes–van Suijlekom theorem, and it is essential to note that it is *not* automatic: it holds only because A_λ is a compression of a convolution (difference-kernel) operator, the structure Step 1 was careful to establish.

The structural hypothesis is essential. The naive statement "a simple isolated even ground state has real-rooted $\hat{\xi}$" is *false in general*: for any positive even compactly supported ξ, the operator $I - P_\xi$ has ξ as simple ground state yet $\hat{\xi}$ need not be real-rooted. The theorem requires the convolution / divided-difference (Carathéodory–Fejér) structure.

Theorem 3.12 (Connes–van Suijlekom 2025)**.** *Let Q be the form on $L^2[-L/2, L/2]$ associated to a real even distribution D via the Schwartz kernel $\widetilde{D}(x-y)$ (i.e. Q is the compression of a convolution operator), lower-bounded with simple isolated even spectral minimum ξ. Then all zeros of $\hat{\xi}(z)$ are real.*

Proof. (i) Carathéodory–Fejér: a Hermitian PSD *Toeplitz* matrix of rank n has its kernel polynomial's zeros on the unit circle — the Toeplitz (difference-kernel) structure is exactly what the convolution hypothesis provides. (ii) Shift Q by a multiple of δ_0 to make it PSD with ξ in its radical; the matrix is of the extended divided-difference class. (iii) Finite truncations converge; Hurwitz passes reality to the limit. ☐

Our A_λ satisfies the hypothesis by Lemma 1.1 (compression of convolution by the even kernel $D_\zeta(|x-y|)$), so the theorem applies with $D = D_\zeta$. *Corollary:* $\hat{\xi}_\lambda$ real-rooted at each λ; RH-neutral (holds for L_{DH}, also of convolution type), so non-circular.

3.5. **Step 4 — Approximation scaffolding (PROVED).** *Role.* Steps 2–3 live entirely at finite λ; to say anything about Ξ (hence about RH) we must control how well the finite-λ world approximates the true completed zeta function. Step 4 supplies the three quantitative estimates the limit in Step 5 consumes: (4.1) that the truncated object Ξ_T is exponentially close to Ξ on the relevant window; (4.2) that the band-limited degrees of freedom suffice to resolve every zero up to the horizon T^* (a Nyquist count, so no zero is missed); and (4.3) that the boundary/amplification losses are negligible. None of these involve RH; they are the classical analytic bookkeeping that makes the reduction rigorous.

4.1. $|\Xi_T(s) - \Xi(s)| \le C\lambda^{5/2+\delta}e^{-\pi\lambda^2}$ on $|t| \le T^*$. **4.2.** $N(T^*) = 2\lambda^2 \log\lambda - \lambda^2 + O(\log\lambda)$ (RvM); DOF $= (\Omega/\pi)(2T^*) = 4\lambda^2\log\lambda$; so $\#\{|\gamma_\rho| \le T^*\}/\text{DOF} = 1 - 1/(2\log\lambda) < 1$. **4.3.** $1 - \omega \le (2/\pi)(y_0/T^*) = O(1/\lambda^2)$ (Boas); amplification $\lambda^{|y|} \le \lambda^{1/2}$ (Phragmén–Lindelöf).

3.6. **Step 5 — The reduction (the jewel) (PROVED).** *This is the step that turns RH into the sign of a number.* Everything before it was preparation: a clean, real-rooted, well-approximating family A_λ. Step 5 now states the payoff. Define the localized minimum

$$\varepsilon_{\text{loc}}(\lambda) = \min_{\|f\|=1,\, f \in E_\lambda} A_\lambda(f, f),$$

the smallest value the Weil form takes on bandwidth-λ test functions. The theorem says RH is *exactly* the statement that this minimum decays slightly faster than $\lambda^{-1/2}$. The mechanism is the one assembled in Steps 2–4: the real-rooted $\hat{\xi}_\lambda$ samples Ξ_T (Step 4.1), its zeros are forced onto the line by real-rootedness (Step 3), the only way the limit can produce an off-line zero of Ξ is a failure of convergence, and by Hurwitz (Step 7) that failure is measured precisely by a non-vanishing deficit $\varepsilon_{\mathrm{loc}}$. The uniform gap of Step 2 (Theorem 3.10) is what guarantees the limit is non-degenerate, so that the "iff" is genuine in both directions.

Theorem 3.13. *With* $\varepsilon_{\mathrm{loc}}(\lambda) = \min_{\|f\|=1, f\in E_\lambda} A_\lambda(f,f)$*, one has* RH $\iff \varepsilon_{\mathrm{loc}}(\lambda) = o(\lambda^{-1/2})$.

Proof. By 4.1 approximate the zeros of Ξ_T; by Steps 2–3 $\hat{\xi}_\lambda$ is real-rooted and its zeros sample Ξ_T; the deficit is $\varepsilon_{\mathrm{loc}}$; by Hurwitz the zeros converge to those of Ξ — both real — iff $\varepsilon_{\mathrm{loc}} = o(\lambda^{-1/2})$. ☐

Reading. After this theorem RH is no longer about zeros: it is the inequality "$\varepsilon_{\mathrm{loc}}$ is small". Step 12.1 will sharpen "small" to the bare sign $\varepsilon_0 \geq 0$. Steps 6–7 guard the reduction (it must reject RH-false inputs and rest on correct limiting logic), and Steps 8–12 ask what kind of object A_λ is and how hard that final sign really is.

3.7. **Step 6 — Boundary fusion and the RH-false witness (PROVED).** *Why a falsifiability test is needed.* A reduction of RH is worthless if it would "prove" RH for an object that is actually RH-false: that would mean the machinery is blind to off-line zeros, i.e. circular or vacuous. Step 6 runs the only available controlled experiment. The Davenport–Heilbronn function Ξ_{DH} shares the functional equation and the same magnitudes as ζ but is *known* to have zeros off the line. We feed it through the same construction and check that it is correctly rejected.

The band edge (carrier $\omega^* \approx \pi$) fuses with the bulk under Hurwitz. For Davenport–Heilbronn the approximants $\hat{\xi}_\lambda$ are real-rooted (Step 3 applies, L_{DH} being of convolution type too) but Ξ_{DH} is not (off-line zero $0.8085 + 85.699i$); by the contrapositive of Hurwitz the convergence $\hat{\xi}_\lambda \to \Xi_{DH}$ *must* fail, and the mechanism is explicit: the off-line zero adds a negative well to the symbol $a(t)$ at $t \approx \gamma_{\mathrm{off}}$, producing $\varepsilon_0 < 0$. So the form does see the violation. This both certifies non-circularity (the test detects RH-false inputs) and localizes *where* a violation would show up (a negative well in the symbol).

3.8. **Step 7 — Hurwitz and Laguerre–Pólya (PROVED).** *The limiting logic, made explicit.* Steps 5–6 invoked "Hurwitz" repeatedly; Step 7 states the two classical theorems they rest on, so the limit is on a rigorous footing. Hurwitz controls when real-rootedness survives a limit; Laguerre–Pólya identifies real-rootedness of Ξ with membership in the class of functions that are transforms of positive-definite forms — which closes the loop back to $A_\lambda \succeq 0$.

Hurwitz: a uniform-on-compacts limit of real-rooted holomorphic functions is real-rooted. $\Xi \in$ LP $\iff$ all zeros real $\iff$ RH; with Lemma 3.3, Ξ is the transform of a PSD form iff $A_\lambda \succeq 0$ $\forall\lambda$.

3.9. **Step 8 — Identification of the limit operator (PROVED) (CCM).** *From a form to an operator.* Steps 1–7 reduced RH to the sign of $\varepsilon_0(\lambda)$ but treated A_λ as an abstract form. To understand the sign one wants to know *what operator* governs the bottom of the spectrum — and here the CCM framework enters: the low spectrum of the rescaled form $B_\lambda = \lambda^2 A_\lambda$ is governed by the Slepian–Pollak prolate operator, and the super-exponentially small eigenvalues live in its near-radical, accessed through the conditioning map $\mathcal{E}$. This identification is what connects our A_λ to the zeta spectral triples of Steps 9–11, and it is RH-neutral: it names the operator without deciding the sign.

$B_\lambda = \lambda^2 A_\lambda$ has low spectrum governed by the Slepian–Pollak prolate operator $W_\lambda\psi(q) = -\partial((\lambda^2 - q^2)\partial)\psi + (2\pi\lambda q)^2\psi$, $|q| < \lambda$, commuting with $P_\lambda \widehat{P}_\lambda P_\lambda$. The tiny eigenvalues of A_λ (e.g. $2.4 \cdot 10^{-48}$ at λ^2=11) arise from the prolate near-radical via $\mathcal{E}(f)(x) = x^{1/2} \sum_{n>0} f(nx)$. **8.1:** carrier $(-1)^m = \chi_m \approx (-1)^m$. Earlier identifications **(NO-GO)** (§2.6).

3.10. **Step 9 — The semilocal prolate operator (COMPUTED, PROOF PENDING).** *Building in the primes.* The bare prolate operator of Step 8 is archimedean; the zeros of ζ require the primes. Step 9 constructs the semilocal operator W_S that incorporates a finite set S of primes through its spectral measure, and — crucially — verifies that the arithmetic it must carry is correct by reconstructing, as *exact identities*, the periodic-orbit (Gutzwiller) data: the prime p is a closed orbit of length $\log p$ and instability weight $p^{-1/2}$. The headline computation (9.5-bis) replaces a previously conjectured Bessel asymptotic by the exact statement that the orbit correction δa_n is a matrix element of the scaling propagator $\cos(LJ)$. All of this is RH-neutral scaffolding: it makes the operator concrete and checks it against the explicit formula, without assuming where the zeros lie.

$W_S = scal^2 + N$ on the cyclic pair $(scal, \xi_S)$; $scal = J_S$ (Jacobi), $N = \mathrm{diag}(n)$, spectral measure $dm_S(s) = |\Gamma_{\mathbb{R}}(\frac{1}{2}+is)|^2 \prod_{p\in S} |1-p^{-(1/2+is)}|^{-2} ds$. Even $\Rightarrow b_n = 0$, $W_S = J_S^2 + \mathrm{diag}(n)$. *The object CCM deferred.*

9.1. $S = \{\infty\}$: $a_n = \sqrt{(n-\frac{1}{2})n}$ reproduces CCM exactly to $n = 15$. **9.2.** $|\delta a_n/a_n| \approx C' p^{-1/2}(n\log p)^{-1/2}$, $C' \approx 0.85$, across $p = 2, \ldots, 23$: $p^{-1/2}$ is the Gutzwiller orbit weight, $(n \log p)^{-1/2}$ the Bessel envelope $J_2(2n\log p)$. *Linear-response identity (validated):* with $w_\infty = |\Gamma_{\mathbb{R}}(\frac{1}{2}+is)|^2$ and the archimedean (Meixner–Pollaczek) orthonormal p_n, $\delta \log a_n = \sum_{p,k} \frac{p^{-k/2}}{k} I_n(k\log p)$, $I_n(L) = \int_{\mathbb{R}} \cos(Ls)(p_n^2 - p_{n-1}^2) w_\infty\, ds$; verified vs. direct δa_n for $p = 23, 29, 31, 37$ (ratio $0.997, 0.997, 1.009, 0.995$, std ≤ 0.05). **9.5-bis (exact closed form, PROVED).** Since w_∞ is the spectral measure of the Jacobi operator J ($a_n = \sqrt{(n-\frac{1}{2})n}$, $b_n = 0$), one has $\int e^{iLs} p_n p_m w_\infty\, ds = \langle p_n, e^{iLJ} p_m\rangle = (e^{iLJ})_{nm}$, hence $\boxed{I_n(L) = [\cos(LJ)]_{n,n} - [\cos(LJ)]_{n-1,n-1}}$. This is exact (verified to 10^{-16} for $L = \log 2, \log 3$, all $n \le 14$). Thus $\delta \log a_n = \sum_{p,k} \frac{p^{-k/2}}{k}\big([\cos(k\log p\, J)]_{n,n} - [\cos(k \log p\, J)]_{n-1,n-1}\big)$: δa_n is the diagonal of the scaling-operator propagator $\cos(LJ)$, $L = \log p^k$ the orbit length — the Gutzwiller picture as an exact identity. **(PROVED, RH-neutral)**

9.5-ter — Fine oscillation law of $I_n(L)$ (PROVED). *The exact $n \to \infty$ asymptotics, closing the fine-oscillation law and superseding the conjectured Bessel form $C_0 J_2(2nL)$.* The matrix J ($a_n = \sqrt{n(n-\frac{1}{2})}$, $b_n = 0$) is the Jacobi matrix of the orthonormal symmetric Meixner–Pollaczek polynomials with $\lambda = \frac{1}{4}$ and spectral measure $w_\infty(s) = |\Gamma_{\mathbb{R}}(\frac{1}{2}+is)|^2$; equivalently $J = 2K_1$ is twice the boost generator of the $su(1,1)$ discrete series $D_{1/4}$ (from $a_n = \sqrt{n(n+2k-1)}$ with $k = \frac{1}{4}$).

Theorem 3.14 (fine oscillation law)**.** *Fix $L > 0$ and set $\omega(L) := 2\arcsin(\tanh L) = 2\,\mathrm{gd}(L)$ (twice the Gudermannian; $\sin(\omega/2) = \tanh L$, $\cos(\omega/2) = \mathrm{sech}\, L$). Then as $n \to \infty$,*

$$[\cos(LJ)]_{n,n} = (\pi n \sinh L)^{-1/2} \cos\big((n+\tfrac{1}{4})\omega(L) - \tfrac{\pi}{4}\big) + O(n^{-3/2}),$$

$$I_n(L) = \frac{2\tanh L}{\sqrt{\pi \sinh L}}\, n^{-1/2} \cos\big((n-\tfrac{1}{4})\omega(L) + \tfrac{\pi}{4}\big) + O(n^{-3/2}).$$

Proof. (i) By the spectral theorem $[\cos(LJ)]_{n,n} = \int_{\mathbb{R}} \cos(Ls) p_n(s)^2 w_\infty(s)\, ds$, real since $b_n = 0$ makes $p_n^2 w_\infty$ even. (ii) In the bulk $|s| < 2a_n$ the Plancherel–Rotach law (Li–Wong; Liouville–Green for the recurrence) gives $p_n\sqrt{w_\infty} = \sqrt{2/(\pi R_n)} \cos\Psi_n + O(n^{-1})$, $R_n = \sqrt{4a_n^2 - s^2}$, $\Psi_n(s) = \sum_{k\le n} \arccos(s/2a_k)$, so $p_n^2 w_\infty = (\pi R_n)^{-1}(1+\cos 2\Psi_n) + \ldots$ (iii) In $\int \cos(Ls)\cos 2\Psi_n\,(\pi R_n)^{-1} ds$ the stationary points solve $2\Psi_n'(s) = \mp L$; with $\Psi_n'(s) = -\sum_k (4a_k^2 - s^2)^{-1/2} \approx -\frac{1}{2}\mathrm{arccosh}(2n/|s|)$ this gives $|s_*| = 2n\,\mathrm{sech}\, L$, and by the envelope theorem the n-frequency is $\omega = \frac{d}{dn} 2\Psi_n(s_*) = 2\arccos(s_*/2a_n) = 2\arccos(\mathrm{sech}\, L) = 2\arcsin(\tanh L)$ (using $1 - \mathrm{sech}^2 L = \tanh^2 L$) — exactly the dispersion, $\neq 2L$, which is why the Bessel form failed. (iv) At s_*, $R_n(s_*) = 2n\tanh L$ and the shift $a_k = k - \frac{1}{4} + O(1/k)$ gives degree $\nu = n - \frac{1}{4}$; equivalently $[\cos(LJ)]_{n,n} = \mathrm{sech}\, L\, P_{n-1/4}(\cos\omega) +$

$O(n^{-3/2})$, and Laplace–Heine $P_\nu(\cos\theta) = \sqrt{2/(\pi\nu\sin\theta)}\cos((\nu+\frac{1}{2})\theta - \frac{\pi}{4}) + O(\nu^{-3/2})$ with $\sin\omega = 2\tanh L \operatorname{sech} L$ yields the stated amplitude $(\pi n \sinh L)^{-1/2}$ and phase. (v) $\cos\Phi_n - \cos\Phi_{n-1} = -2\sin(\omega/2)\sin(\Phi_n - \omega/2)$ with $\sin(\omega/2) = \tanh L$ gives I_n. □

Validation (exact engine, $N \gg n$): the law matches the computed $[\cos(LJ)]_{n,n}$ and I_n to relative error $\leq 3{\cdot}10^{-4}$ (decreasing as $O(n^{-1})$), uniformly over orbit lengths $L = \log 2, \log 3, \log 4, \log 5, \log 8$, $n \leq 2500$; the fitted dispersion matches $2\arcsin(\tanh L)$ to 4 digits on $L \in [0.3, 2.2]$ and $B_n\sqrt{\pi n \sinh L} = 1$ to 4 digits independently of L. This **refutes $C_0 J_2(2nL)$ for a precise reason** (the true frequency is the Gudermannian $2\arcsin(\tanh L)$, not $2L$; the earlier "envelope grows" refutation observed the pre-asymptotic $n \leq 80$ regime). The Gudermannian dispersion is the $su(1,1)$ signature: the circular angle ω is conjugate to the hyperbolic orbit rapidity L. **(PROVED — Plancherel–Rotach + stationary phase / Laplace–Heine; constants verified to 4 digits.) 9.3.** Primes compress the moments: $\mu_2/\mu_0 = 0.50 \to 0.16 \to 0.08 \to 0.05$. **9.4.** $a_n^S \sim n \Rightarrow$ Carleman diverges $\Rightarrow J_S$ essentially self-adjoint; the negative eigenvalues (zeros) live in the orthogonal complement (Sonin). **(PROVED)**

3.11. **Step 10 — Counting and periodic orbits (PROVED)/(COMPUTED, PROOF PENDING).** *Confirming the operator is the right one.* An operator proposed to carry the zeros must reproduce the zero-counting function of ζ, smooth part and fluctuation alike. Step 10 is that consistency check: the smooth count matches the CCM semiclassical law, and — the decisive point — the fluctuation $S(E) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+iE)$ is reproduced *exactly* as a Gutzwiller sum over prime orbits, numerically to three digits. Together with Step 9 this fixes the arithmetic of the operator beyond reasonable doubt; what remains undecided is only the one thing that is RH (the sign / the reality of the negative spectrum), handled in Steps 11–12.

10.1. $N(E) = N_{\text{smooth}}(E) + S(E)$, $N_{\text{smooth}} = \frac{E}{2\pi}(\log\frac{E}{2\pi} - 1) + \frac{7}{8}$, $S(E) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+iE)$; CCM's $\sigma(E,\lambda)$ reproduces N_{smooth}. **10.2.** $S(E) = -\frac{1}{\pi}\sum_{p,k}(kp^{k/2})^{-1}\sin(kE\log p)$; verified $S_{\text{real}} = S_{\text{primes}}$ (-0.293 vs -0.294 at E=40). A Gutzwiller sum: orbit $= p$, length $\log p$, weight $p^{-1/2}$, repetitions k. **(PROVED)10.3.** $H_\lambda = (p^2-\lambda^2)(q^2-\lambda^2)$ has open (scattering) orbits $\Rightarrow$ only N_{smooth}; the closed orbits $\log p$ come from the finite places.

3.12. **Step 11 — Sonin space and de Branges function (OPEN)→(WALL = RH).** *Where the zeros actually live, and why this is the wall.* The positive spectrum built in Steps 8–10 is archimedean scaffolding; the nontrivial zeros sit in the *negative* spectrum of the self-adjoint prolate operator, carried by the Sonin space. Step 11 constructs this negative spectrum (rigorously, via a monotone Prüfer flow) and validates its counting against the semiclassical law, then asks the RH question in de Branges language: are the associated γ_n real? Every attempt to force this by a *positivity* (Hermite–Biehler, a scalar de Branges quotient, a positive Hamiltonian $H_S \succeq 0$) is refuted with its reason — the scalar positivity is false for ζ (Conrey–Li/Sarnak), and the correct object is an indefinite, finite-Pontryagin-index transfer matrix. The reality of the γ_n is reached here from yet another independent direction and remains undecided: *this is the same wall*, now wearing its de Branges costume.

$S_\lambda = \{f \in L^2(\mathbb{R})^{ev} : f = 0, \hat{f} = 0 \text{ on } [-\lambda,\lambda]\}$ carries the negative spectrum of W_{sa} (zeros, $\sim -\gamma_n^2$ UV). **11.1.** $E_S(z) = 1/\Lambda_S(\frac{1}{2} - iz)$ entire, $|E_S(x)|^{-2} = dm_S(x)$ (verified). **11.2.** E_S *not* Hermite–Biehler: at $z = 2+i$, $|E_S(z)| = 5.13 < |E_S(\bar{z})| = 5.80$ (gamma factor decays in the upper half-plane). **(NO-GO)11.2-bis (structural no-go).** The scalar $\Theta_S(z) = \Lambda_S(\frac{1}{2}-iz)/\Lambda_S(\frac{1}{2}+iz)$ has poles/zeros at $z = 2\pi m/\log p \pm i/2 \in \mathbb{C}_+$, hence is not Schur/contractive in $\mathbb{C}_+$: it cannot be a characteristic function of a self-adjoint extension nor a de Branges quotient. Any scalar Robin condition from Λ_S inherits these singularities, and (DH sharing the partial-Euler structure) is exactly what marginality (12.2) rules out. *The right finite-S object is a 2×2 J-unitary transfer matrix* (Tate), giving a de Branges canonical system $J\dot{Y} = zH_SY$; the wall is $H_S \succeq 0$. **(NO-GO)** (sharpened → §4.B) **11.3.** HB = reality of zeros requires $s \leftrightarrow 1-s$. $\Lambda(s) = \Lambda(1-s)$ makes $\xi(\frac{1}{2}+iz)$ real, even, zeros γ_n (verified $\operatorname{Im}\xi(\frac{1}{2}+it) \sim 10^{-23}$). The partial Λ_S has no functional equation: its zeros are trivial (neg. imag. axis) and Euler ($\operatorname{Im} z = -\frac{1}{2}$), not γ_n. **(PROVED)11.3-bis (Sonin negative spectrum — constructed and validated).** By CCM (Thm 2.6.iv) the negative spectrum of W_{sa} is that of $W_\lambda\psi = -\partial((\lambda^2 - q^2)\partial)\psi + (2\pi\lambda q)^2\psi$ on $L^2(\lambda, \infty)$ with BC (16) regular at $q = \lambda$ and (17) at $q = \infty$. With $u = R\sin\theta$, $(q^2 - \lambda^2)u' = R\cos\theta$, the equation $((q^2 - \lambda^2)u')' + ((2\pi\lambda q)^2 - \xi)u = 0$ gives the *monotone* Prüfer flow $\theta' = \cos^2\theta/(q^2 - \lambda^2) + ((2\pi\lambda q)^2 - \xi)\sin^2\theta$, $\theta(\lambda) = \pi/2$. For $\xi < 0$ both terms are positive, so θ increases in q and in $-\xi$: every eigenvalue is a crossing $\theta = m\pi$, none skipped or spurious — reliable even at the limit-circle oscillatory endpoint where the asymptotic functional (17) (valid only for $x \gg \mu/12$) fails. *Validation:* the counting $N(E) = \#\{-\xi \leq E^2\}$ has spectral density dN/dE matching the CCM law $d(2\sigma)/dE$ to ≈ 0.9 uniformly across $\lambda = 1, 2, 3$ (increments $\Delta N/\Delta(2\sigma) = 0.85$–$0.96$, no λ-bias); the cumulative count correctly gives zero eigenvalues below the first ($\lambda = 3$: first at $E \approx 16$). **(PROVED)** (reliable construction; density validated) **11.4.** Closing HB with the functional equation = γ_n real. **(WALL = RH)**

3.13. **Step 12 — The wall (WALL = RH).** *The single statement that is RH.* Step 12 names the wall in its cleanest form and proves the one structural fact that explains why sixty phases failed to climb it. By 12.1 the entire chain collapses to the positivity $(P) : \varepsilon_0(\lambda) \geq 0$ for all λ, which is RH. The marginality theorem (12.2) then shows why (P) is so hard: the negative central well of the symbol and the band resolution are in a fixed, scale-invariant ratio $\sqrt{2}/\pi$ that depends only on the *magnitudes* $|\Lambda(n)|$ — and Davenport–Heilbronn has the very same magnitudes yet $\varepsilon_0 < 0$. Hence no magnitude/size method can ever decide the sign of ε_0: the sign is a *phase* invariant. This is the quantitative core of the program's negative discovery, and it tells any future prover exactly what kind of argument is required — one that is phase-sensitive and encodes the location of the zeros by construction, which is precisely the CCM operator route whose stability rests on the new Lemma 3.9.

12.1. $(P) : \varepsilon_0(\lambda) \geq 0\ \forall\lambda \iff$ RH (Lemma 3.3 and DH). **12.2 (marginality).** Central well half-width $\delta_\lambda = (2|a(0)|/a''(0))^{1/2}$, resolution $r_\lambda = \pi/\Omega$; with $P_\lambda \sim 2\lambda$, $M_2 \sim 8\lambda\log^2\lambda$ (PNT), $a(0) \sim -2\lambda$, $a''(0) \sim M_2$: $\frac{2\delta_\lambda}{r_\lambda} = \frac{\sqrt{2}}{\pi} \approx 0.45$. Since $\delta_\lambda, r_\lambda$ depend only on $\{|\Lambda(n)|\}$, no magnitude functional decides $\operatorname{sign}\varepsilon_0$ (DH, same magnitudes, $\varepsilon_0 < 0$). The sign is a phase invariant — the quantitative form of MW-4 (wrong sign) and MW-2 (phase unreachable by magnitude). **(PROVED)**

4. Future work

4.A — Closed. The law of δa_n (Step 9.5-bis) is now exact: $I_n(L) = [\cos(LJ)]_{n,n} - [\cos(LJ)]_{n-1,n-1}$, a matrix element of the scaling-operator propagator, superseding the (numerically refuted) Bessel/DIK conjecture. No RH-neutral residue remains in Step 9.

4.B — The frontier: adelic transfer matrices and canonical-system positivity. By 11.2-bis the right finite-S object is not the scalar partial product but a 2×2 J-unitary transfer matrix from Tate's local functional equation: take $\varphi = \varphi_\infty \otimes \bigotimes_p \varphi_p$ on $\mathbb{A}_\mathbb{Q}$, $\varphi_p = \mathbf{1}_{\mathbb{Z}_p}$ for $p \in S$, and keep both $Z_p(\varphi_p, s)$ and the dual $Z_p(\widehat{\varphi}_p, 1-s)$ as entries of $T_p(z) \in SU(1,1)$. Then $T_S = T_\infty \prod_{p \in S} T_p$ is the transfer matrix of a de Branges canonical system $J\dot{Y} = zH_S Y$ (this is why scalar HB fails: for $p \notin S$, $\widehat{\varphi}_p \neq \varphi_p$, and the dual side restores the discarded data). Three linked targets: (1) *Boundary triple (RH-neutral, doable):* W_λ on $|q| < \lambda$ has limit-circle log endpoints, deficiency $(4,4)$, 2 even; tabulate the matrix Weyl function $M_\lambda(z)$, $\operatorname{Spec}_{\text{disc}}(W_{\lambda,B}) = \{E : \det(B - M_\lambda(E)) = 0\}$; classify the Fourier-invariant B and compare B_{CCM} with Burnol's K_λ/Z_λ. (2) *The Hamiltonian:* prove T_p J-unitary on $\mathbb{R}$, J-contractive in $\mathbb{C}_+$, so $H_S \succeq 0$ and E_S is HB; Hurwitz closes $S \to \mathbb{P}$. *The wall is $H_S \succeq 0$* (compare carefully with Conrey–Li IMRN 2000). (3) *Dynamical symmetry:* $s \leftrightarrow 1-s$

emerges from the $\widehat{\mathbb{Z}}^\times$ symmetry of Bost–Connes KMS states at $\beta = 1$; the semilocal Sonin operator should be a restriction of a Bost–Connes-type Hamiltonian. *Decisive RH-neutral test:* can the boundary condition at $q = \pm\lambda$ be stated without reference to the zeros? If not, that is where RH is inserted.

4.C — The wall. $(P) : \varepsilon_0 \geq 0$ is RH (MW-1), immune to magnitude methods (Step 12.2 = MW-4/MW-2 quantitative). The collapse theorem says any unconditional route must be phase-sensitive and encode the zeros by construction — the CCM/Hilbert–Pólya route of 4.B, the only non-circular candidate, not closed.

Summary of status. Steps 1–10 are proved or constructed-and-validated. The sixty-phase unconditional results: the localized Weil detector (Arc B, Fronts A, B); the reduction RH $\iff$ $\varepsilon_{\text{loc}} = o(\lambda^{-1/2})$; the structure theorem; real-rootedness (CvS); the approximation scaffolding; the marginality theorem; and the construction and validation of the semilocal prolate operator with its periodic-orbit weight. Step 9.5 is now proved (Theorem 3.14, the Gudermannian fine oscillation law); Step 11 is open and, with Step 12, is the wall, reached identically from primes, zeros, operator, de Branges, and the seven structural walls of the earlier arcs: the reality of the nontrivial zeros, i.e. RH. The contribution is a rigorous localization of RH onto a single phase-sensitive statement, the elimination — with precise obstructions — of every alternative the program could construct or find in the literature, the CCM scaffolding around the wall, and the precise identification of the missing ingredient.

References

[1] A. Connes, C. Consani, H. Moscovici, *Zeta spectral triples*, arXiv:2511.22755 (2025).
[2] A. Connes, W. D. van Suijlekom, *Quadratic Forms, Real Zeros and Echoes of the Spectral Action*, arXiv:2511.23257 (2025).
[3] A. Connes, C. Consani, *Spectral triples and ζ-cycles*, arXiv:2106.01715 (2021).
[4] A. Connes, H. Moscovici, *Prolate spheroidal operator and Zeta*, arXiv:2112.05500 (2021).
[5] A. Connes, H. Moscovici, *Zeta zeros and prolate wave operators*, arXiv:2310.18423 (2024).
[6] A. Connes, C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. 27 (2021).
[7] P. Kurasov, P. Sarnak, *Stable polynomials and crystalline measures* (2020).
[8] H. L. Montgomery, *The pair correlation of zeros of the zeta function* (1973).